

Lab 3: Exploratory Data Analysis (EDA)

Objective

To perform exploratory data analysis (EDA) on a dataset using Python libraries such as `pandas`, `matplotlib`, and `seaborn`, and to interpret the key patterns, trends, and relationships between variables.

Theory

Exploratory Data Analysis is the process of visually and quantitatively summarizing data to identify patterns, anomalies, and insights before model building. It provides an intuitive understanding of the dataset, highlighting distributions, correlations, and possible outliers.

Common EDA Tasks:

- Summarizing data using descriptive statistics.
- Visualizing feature distributions and relationships.
- Detecting correlations and multicollinearity.
- Identifying missing or abnormal data patterns.

Major Functions Used

- `df.describe()` – Basic statistical summary.
- `df.corr()` – Pairwise correlation matrix.
- `sns.heatmap()` – Visual correlation map.
- `sns.histplot()`, `sns.boxplot()`, `sns.pairplot()` – Univariate and multivariate visualization.

Procedure

1. Import Libraries and Load Data

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 # Load cleaned dataset from previous lab
7 df = pd.read_csv("data/cleaned_data.csv")
8
9 print("Shape:", df.shape)
10 print(df.head())
```

2. Basic Statistical Summary

```
1 # Descriptive statistics for numerical features
2 print(df.describe())
3
4 # Information about dataset
5 print(df.info())
6
7 # Unique values per column
8 print(df.nunique())
```

Explanation: This step provides mean, median, standard deviation, and count, helping you quickly identify feature ranges and missing or abnormal values.

3. Visualize Distributions

```
1 # Histogram and density plot
2 sns.histplot(df['Age'], kde=True)
3 plt.title("Distribution of Age")
4 plt.show()
5
6 # Boxplot for salary to detect outliers
7 sns.boxplot(x=df['Salary'])
8 plt.title("Salary Distribution and Outliers")
9 plt.show()
```

Observation: Histograms reveal shape (normal, skewed, bimodal). Boxplots highlight median, quartiles, and extreme values (outliers).

4. Relationship Between Variables

```
1 # Pairwise relationships among numeric features
2 sns.pairplot(df.select_dtypes('number'))
3 plt.suptitle("Pairwise Relationships", y=1.02)
4 plt.show()
```

Explanation: Pairplots show relationships and correlations between all numeric columns. Clusters or linear patterns often indicate potential predictive relationships.

5. Correlation Heatmap

```
1 corr_matrix = df.corr(numeric_only=True)
2 plt.figure(figsize=(8,6))
3 sns.heatmap(corr_matrix, annot=True, cmap='coolwarm',
4             linewidths=0.5)
5 plt.title("Correlation Heatmap")
6 plt.show()
```

Observation: Correlation coefficients close to +1 or -1 show strong linear relationships. High correlation among features may cause multicollinearity in models.

6. Categorical Variable Analysis

```
1 # Count plot for categorical features
2 sns.countplot(x='Gender', data=df)
3 plt.title("Gender Distribution")
4 plt.show()
5
6 # Cross-tabulation: Gender vs City
7 pd.crosstab(df['Gender'], df['City'], normalize='index') * 100
```

Explanation: Count plots show category frequencies. Cross-tabulation (or contingency tables) helps understand interaction between two categorical variables.

7. Outlier Detection (Z-Score Method)

```
1 from scipy import stats
2
3 z = np.abs(stats.zscore(df.select_dtypes('number')))
4 outliers = (z > 3).sum(axis=0)
5 print("Outliers per column:\n", outliers)
```

Explanation: Z-score method detects extreme values beyond three standard deviations, helping decide if they should be removed or adjusted.

8. Full EDA Script

```
1 import pandas as pd
2 import seaborn as sns
3 import matplotlib.pyplot as plt
4
5 df = pd.read_csv("data/cleaned_data.csv")
6
7 # Summary statistics
8 print(df.describe())
9
10 # Correlation heatmap
11 corr = df.corr(numeric_only=True)
12 sns.heatmap(corr, annot=True, cmap='coolwarm')
```

```
13 plt.show()
14
15 # Histograms and pairplots
16 sns.histplot(df['Age'], kde=True)
17 plt.show()
18
19 sns.pairplot(df.select_dtypes('number'))
20 plt.show()
```

Listing 2: Complete EDA Workflow

Discussion

EDA reveals the structure and distribution of data, outliers, and relationships between features. Understanding these aspects is critical for selecting appropriate preprocessing and modeling techniques. Visual analysis often exposes issues invisible in numerical summaries.

Exercises

1. Load a dataset (e.g., Titanic, Iris, or any Kaggle dataset).
2. Compute descriptive statistics and interpret results.
3. Create histograms, boxplots, and heatmaps for key variables.
4. Identify two strong and two weak correlations.
5. Write short interpretations of the discovered relationships.

Outcome

Students will be able to perform descriptive and visual analyses, recognize patterns and outliers, and draw meaningful inferences from raw datasets — preparing them for feature engineering and model development in later labs.